Hearing Drones in 3D: Convolutional Detection and Direction-Finding from a Microphone Array

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Project Category: Applied – Audio / Convolutional Networks / Defense & Security

Problem Definition

Drones are hard to see and often fly with no radio link, defeating cameras and radio detectors. Sound is a cheap fallback: rotors are loud and distinctive. From a ground microphone array we want a network reporting ten times per second whether a drone is audible and where: azimuth (compass bearing) and elevation (angle above the horizon).

Key Challenges

A distant drone is quiet and easily confused with wind, traffic, and aircraft. It fills a tiny fraction of the search space, so "nothing" scores well – severe class imbalance. Recordings are continuous: neighbouring windows look almost identical, so a random split would leak training data into test.

Data

Western Sydney University's *UAV Acoustic Localization Dataset* (CC BY-SA 4.0) downloads freely: ~75 minutes of a DJI Air 3 at two sites, ~10 minutes of drone-free background, with GPS logs every 100 ms giving direction and speed: labels come free. The 24 microphones sit on a 1.3 m tripod wired as four synchronised six-channel recorders. A downloaded slice confirmed labels and geometry agree; we train on one site, test on another.

Proposed Method

From sound to images: Sound from one direction reaches the microphones at slightly different times. Delaying each by those offsets and summing reinforces a correct guess and partly cancels a wrong one. Sweeping the dataset's direction grid turns each 100 ms of audio into a 91×23 image[1] of energy per direction; the drone is a bright blob. Fixed signal processing makes this an image problem.

Models: The baseline picks the brightest pixel, as classical systems do, failing on noise and at range. We train, from scratch, an encoder–decoder CNN outputting a per-pixel drone probability plus one "drone present" score, with class-weighted cross-entropy and Adam. Consecutive images enter as extra input channels.

Evaluation

On the unseen test site: precision–recall and false alarms per hour for detection; median angular error and hit rates at the 10° the dataset calls correct. Everything is repeated on six-microphone subsets – a single recorder, then six spanning the full 1.3 m – and for several numbers of stacked images.

Related Work and Contribution

Rodríguez et al. (2026) released this dataset and a single-frame model, naming temporal context as future work and publishing no code; Al-Emadi et al. (2019) detect from one microphone, without direction. Our question is practical: how little array does this task need? Six of the 24 microphones already span the whole 1.3 m, so the other 18 buy no angular resolution – only suppression of the false peaks a sparse array invents. We hypothesise that temporal context supplies it instead: a real drone persists while false peaks flicker – making cheap, widely spaced arrays operationally practical. We also re-test their reported drop in false alarms over classical beamforming: they scored it by output stability; our drone/no-drone score gives a true false-alarm rate. As an extra we try speed from an approaching drone's Doppler pitch shift, though throttle moves rotor pitch more than motion.

[1] Azimuth $-180°$ to $180°$ and elevation $0°$ to $88°$, both in $4°$ steps, giving 91×23 directions. $4°$ is finer than the array's $\approx 7°$ beamwidth at 2200 Hz, so the main lobe is oversampled rather than missed.

References

1. B. J. Rodríguez, S. F. Chevtchenko, M. Herrera Martínez, Y. Bethi, S. Afshar. *Acoustic Imaging for UAV Detection: Dense Beamformed Energy Maps and U-Net SELD.* arXiv:2508.00307, Aug 2025 (rev. Jul 2026); IWAENC 2026.
2. B. J. Rodríguez, S. F. Chevtchenko, S. Afshar. *UAV Acoustic Localization Dataset: 24-Channel Beamformed Recordings of a DJI Air 3 Drone with Synchronized GPS Flight Logs.* Western Sydney University, published 15 Jul 2026. DOI 10.26183/07nm-4p35.
3. S. Al-Emadi, A. Al-Ali, A. Mohammad, A. Al-Ali. *Audio Based Drone Detection and Identification using Deep Learning.* IWCMC, 2019. DOI 10.1109/IWCMC.2019.8766732.